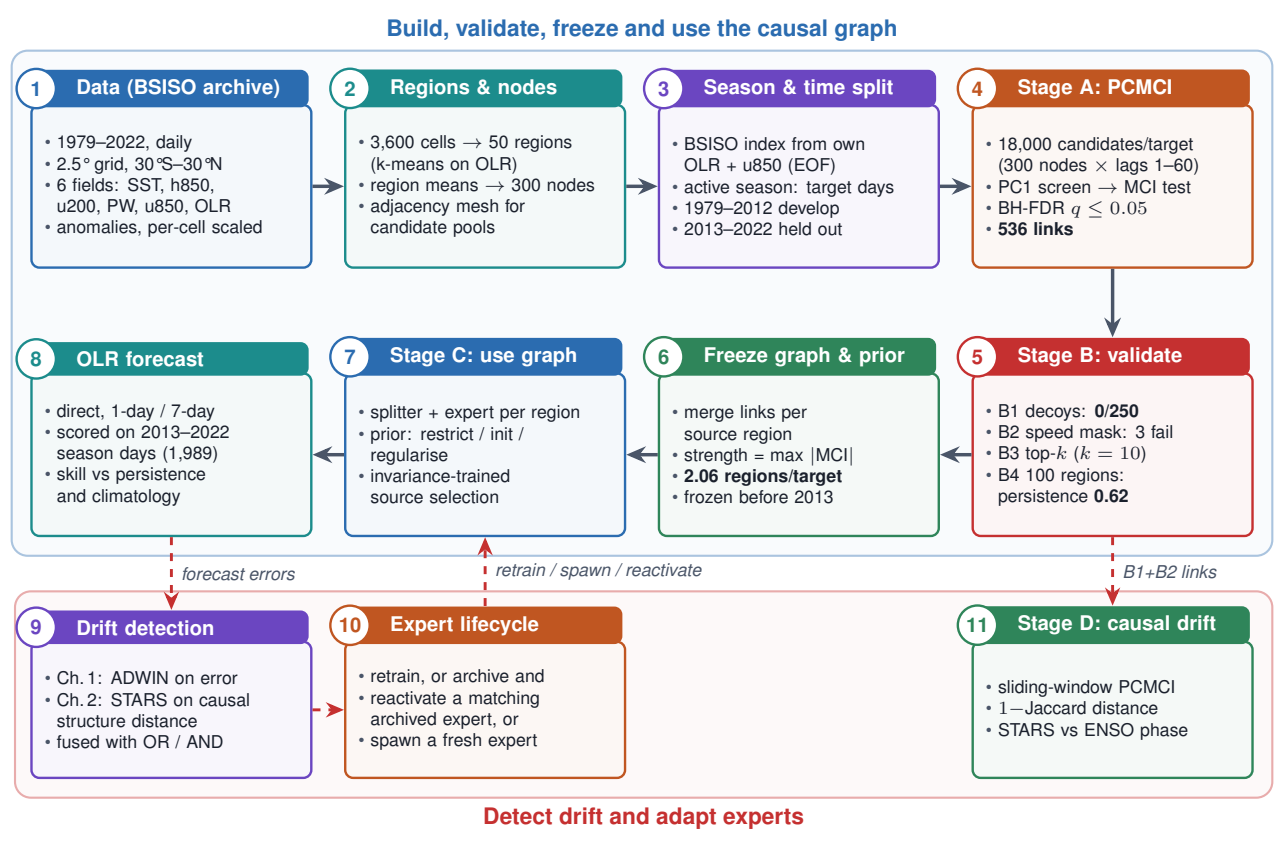

Build, validate, freeze and use the causal graph
1
Data (BSISO archive)
• 1979–2022, daily
• 2.5° grid, 30°S–30°N
• 6 fields: SST, h850, u200, PW, u850, OLR
• anomalies, per-cell scaled
2
Regions & nodes
• 3,600 cells → 50 regions (k-means on OLR)
• region means → 300 nodes
• adjacency mesh for candidate pools
3
Season & time split
• BSISO index from own OLR + u850 (EOF)
• active season: target days
• 1979–2012 develop
• 2013–2022 held out
4
Stage A: PCMCI
• 18,000 candidates/target (300 nodes × lags 1–60)
• PC1 screen → MCI test
• BH-FDR $q \leq 0.05$
• 536 links
5
Stage B: validate
• B1 decoys: 0/250
• B2 speed mask: 3 fail
• B3 top-k ($k = 10$)
• B4 100 regions: persistence 0.62
6
Freeze graph & prior
• merge links per source region
• strength = max |MCI|
• 2.06 regions/target
• frozen before 2013
7
Stage C: use graph
• splitter + expert per region
• prior: restrict / init / regularise
• invariance-trained source selection
8
OLR forecast
• direct, 1-day / 7-day
• scored on 2013–2022 season days (1,989)
• skill vs persistence and climatology
forecast errors
retrain / spawn / reactivate
B1+B2 links
9
Drift detection
• Ch. 1: ADWIN on error
• Ch. 2: STARS on causal structure distance
• fused with OR / AND
10
Expert lifecycle
• retrain, or archive and
• reactivate a matching archived expert, or
• spawn a fresh expert
11
Stage D: causal drift
• sliding-window PCMCI
• 1−Jaccard distance
• STARS vs ENSO phase
Detect drift and adapt experts